

Bacteria, fungi and parasites as agents of cancer

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Introduction

Microbes and cancer

- Cancer is one of the most aggressive and deadly diseases.
- It is a multifactorial disease.
- Genetic alterations play a key role.
- Environmental factors also contribute to its development and progression.
- Infections induced by certain viruses, bacteria, fungi and parasites constitute risk factors for cancer.
- Susceptibility to infectious diseases is higher in cancer patients.
- Globally, **15% of cancers** are a result of infection with oncogenic pathogens.

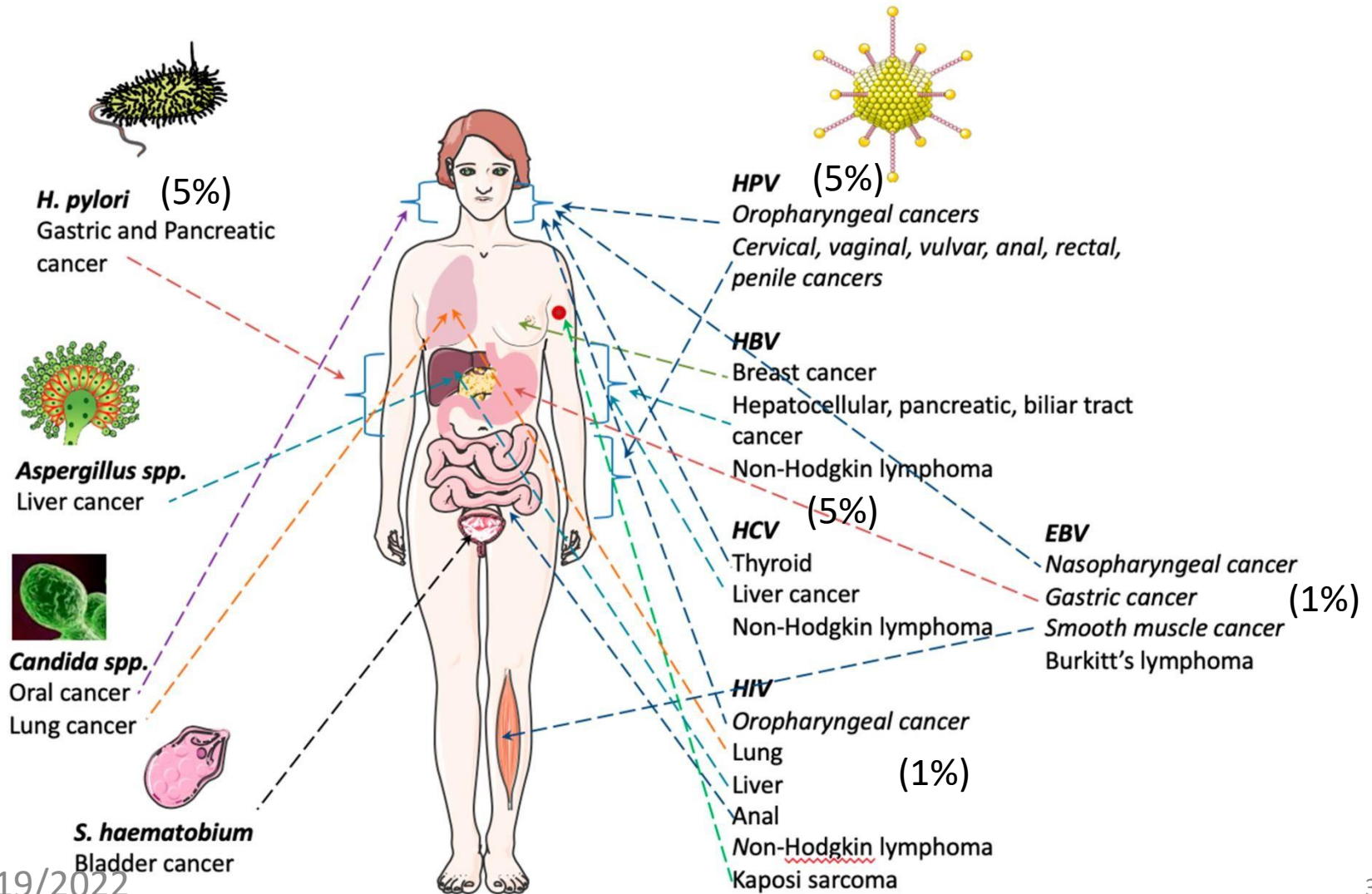
Introduction

Microbes and cancer

- 10 species have been identified by the International Agency for Cancer Research (IARC) as carcinogenic agents to humans.
 - **Clonorchis sinensis** (infection with (iw)), also known as the Chinese liver fluke
 - **Epstein-Barr virus (EBV)** (iw)
 - ***Helicobacter pylori*** (iw)
 - **Hepatitis B virus** (chronic (iw))
 - **Hepatitis C virus** (chronic (iw))
 - **Human immunodeficiency virus type 1 (HIV-1)** (iw)
 - **Human papilloma virus (HPV)** types 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59 (iw)
 - **Human T-cell lymphotropic virus type I (HTLV-1)** (iw)
 - ***Opisthorchis viverrini*** (iw), also known as the Southeast Asian liver fluke
 - ***Schistosoma haematobium*** (iw)

Introduction

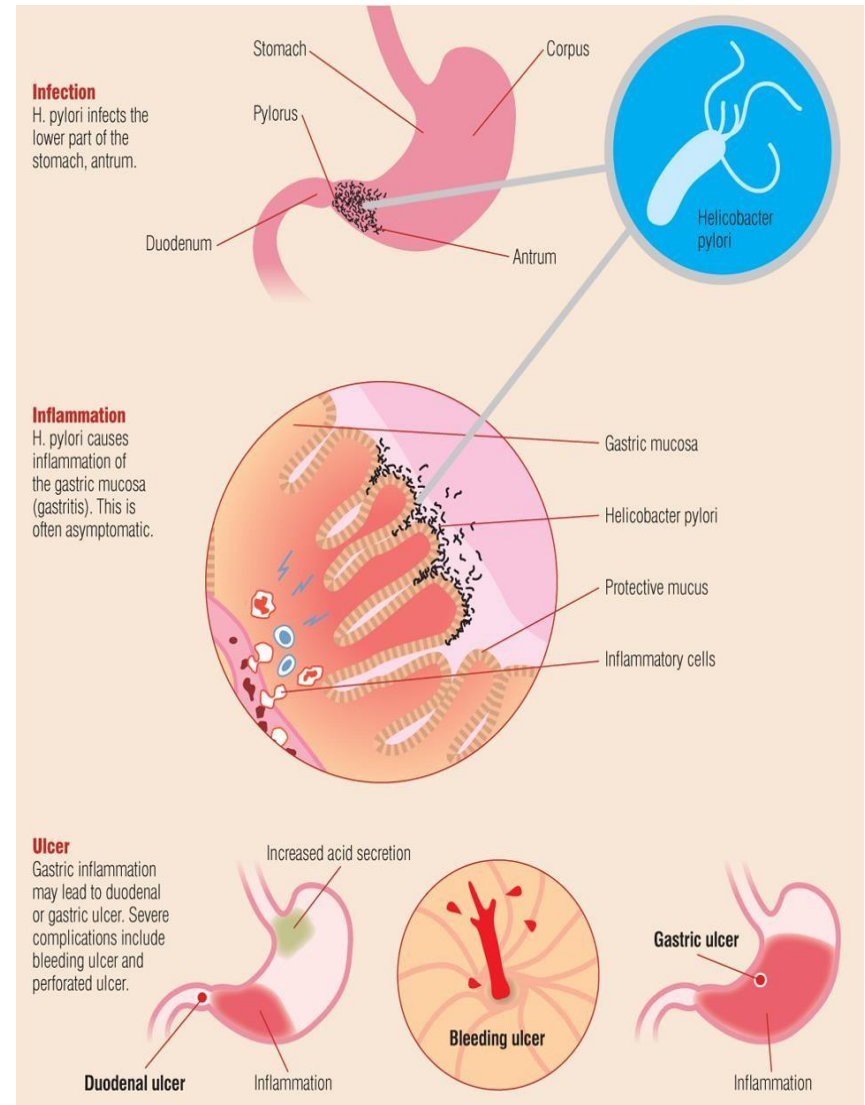
Microbes and cancer



Helicobacter pylori

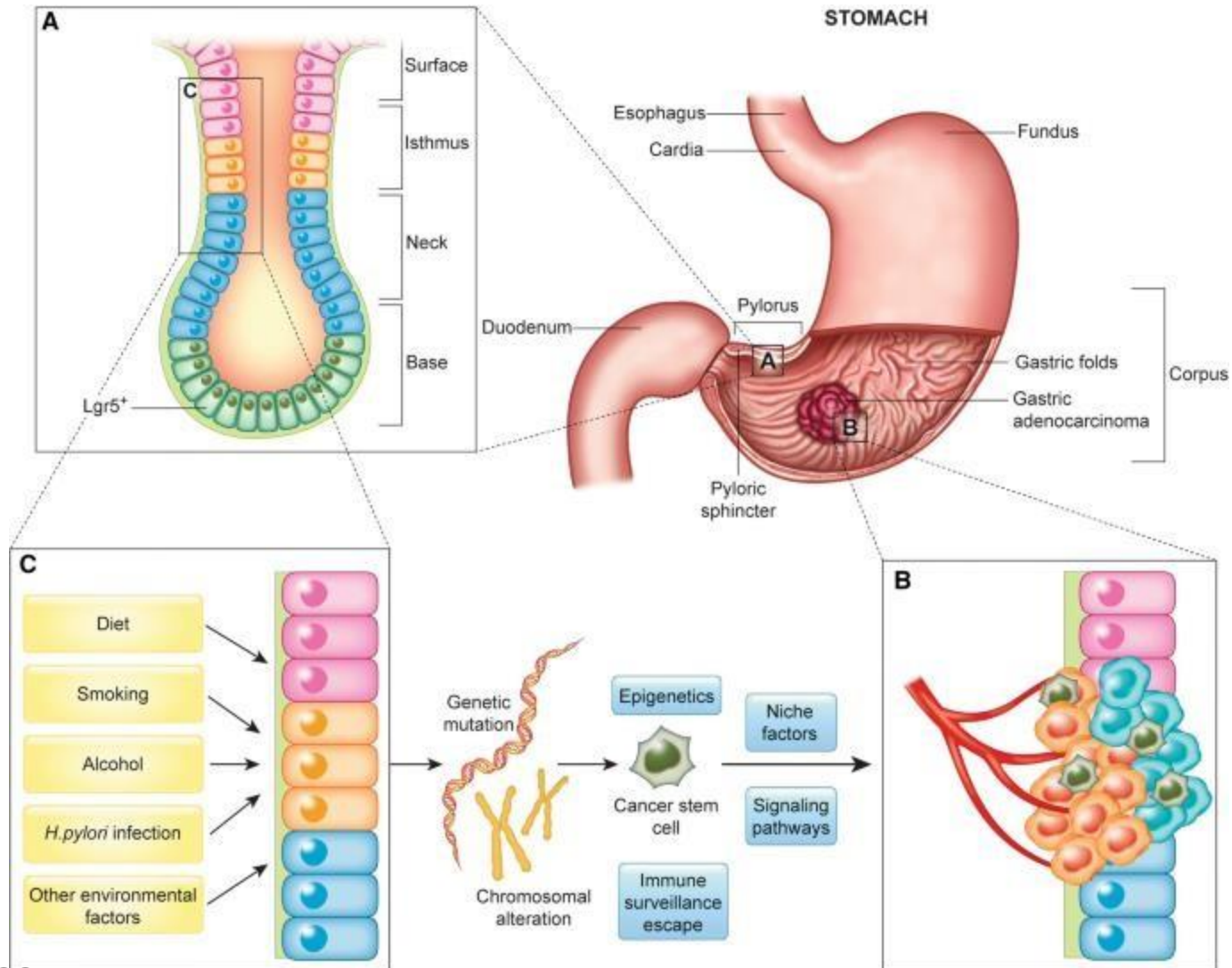
Infection..Inflammation..Adenocarcinoma

- *H. pylori* infects more than 50% of the world's population.
- Infections are typically developed during childhood and usually remain latent for decades.
- Is able to adapt to the extreme acidic conditions of the stomach.
- It infects the lower part of the stomach, antrum.
- Causes inflammation of the gastric mucosa which is often asymptomatic.
- The inflammation may lead to duodenal or gastric cancer.



Bacterial carcinogens

Gastric Adenocarcinoma



Bacterial carcinogens

Helicobacter pylori infection and adenocarcinoma

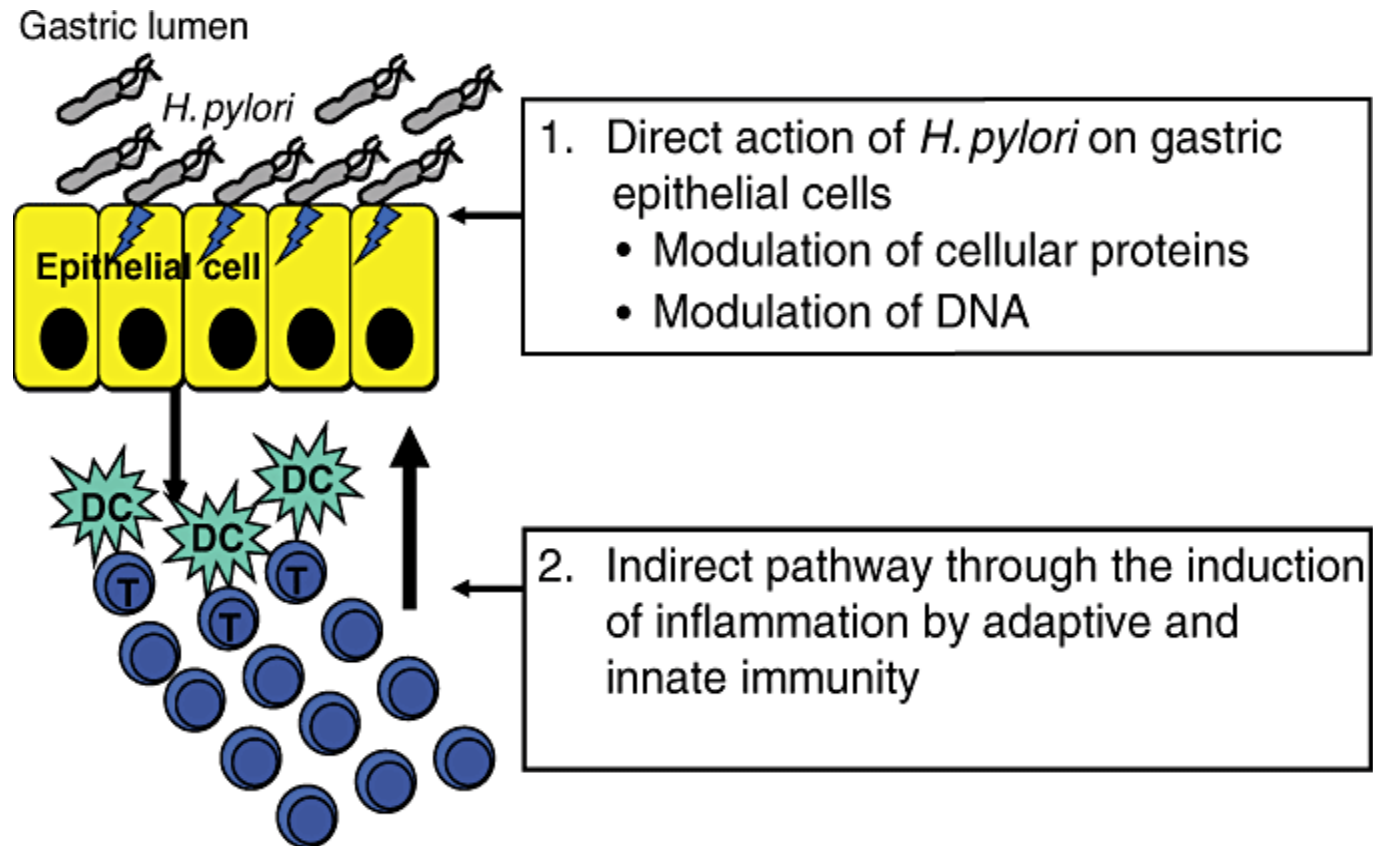


Fig. Mechanism for gastric cancer development by *Helicobacter pylori* infection

Helicobacter pylori infection and adenocarcinoma

Direct effects

- *H. pylori* can have direct effects on the molecular make-up of the gastric epithelial cells through the toxic action of **virulence factors**.
- Changes that lead to malignant transformation include:
 - Mutations of cell-cycle regulating genes,
 - Deficiencies in DNA repair mechanisms,
 - Loss of a cell's adhesive properties
- Two extensively researched virulence factors are:
 - Cytotoxin-associated gene A (CagA) and
 - Vacuolating cytotoxin A (VacA).
- Are associated with precancerous gastric lesions and progression to a malignant phenotype.

Helicobacter pylori infection and adenocarcinoma

Direct effects - *H. pylori* induced intracellular signalling

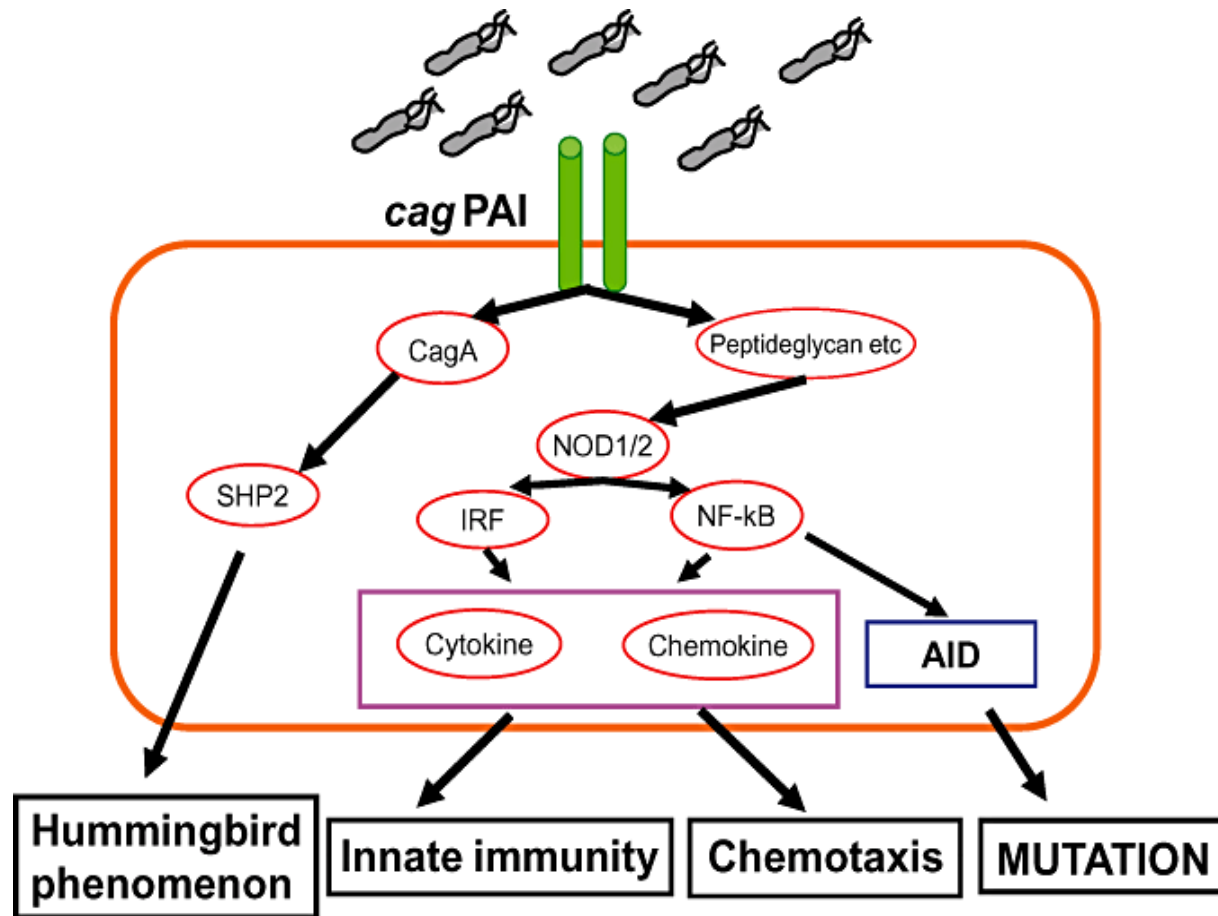


Fig. *Helicobacter pylori*-induced intracellular signaling events that are involved in gastric cancer development.

Helicobacter pylori infection and adenocarcinoma

Indirect effects through inflammatory processes

- *H. pylori* infection mounts a chronic inflammatory response resulting in an increased cell turnover – **accumulation of mitotic errors**.
- Persistent inflammation of the corpus by *H. pylori* results in **atrophic gastritis**, a risk factor for GC.
- Atrophic gastritis leads to a rise in pH and hypochlorhydria or achlorhydria.
- This alkaline environment facilitates the colonization and proliferation of *H. pylori*.
- *H. pylori*-gastritis is predominantly caused by a CD4 Th1-cell response.
- Neutrophils and macrophages are recruited which produce an excessive amount of reactive oxygen and reactive nitrogen species (ROS/RNS).

Helicobacter pylori infection and adenocarcinoma

Indirect effects through inflammatory processes

- ROS, along with the superoxide and the hydroxyl ion produced by *H. pylori*, result in an **increased oxidative stress and DNA damage**.
- **Persistent infection** by *H. pylori* is achieved through a **variety of mechanisms**.
 - *H. pylori* can protect itself from toxic substances such as oxidative species.
 - *H. pylori* can induce the apoptosis of macrophages and
 - It can increase the expression of proinflammatory factors.
- Gastric mucosal expression of **multiple cytokines** (IL1B, IL-6, IL-8 and TNF α) and cyclooxygenase-2 (COX-2) is increased in the nucleus of the mucosal cells.
- This accelerates the progression of atrophic changes and induces intracellular signalling transformation.

Fungal carcinogens

Introduction

- Some fungal infections are thought to be associated with a higher risk of developing cancer.
- Some strains of filamentous fungi, including
 - *Aspergillus flavus*, *Aspergillus parasiticus* and *Aspergillus nomius*, produce aflatoxins able to induce liver cancer.
- Candida infections has been demonstrated in lung cancer.
- Candida infections are also a risk factor for development of oral carcinoma.

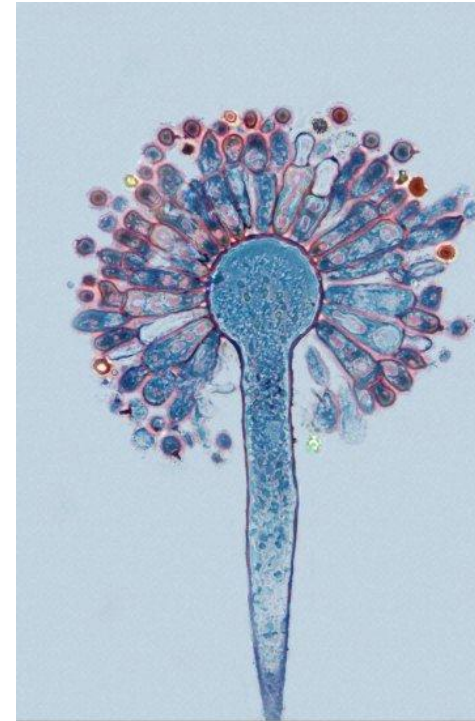
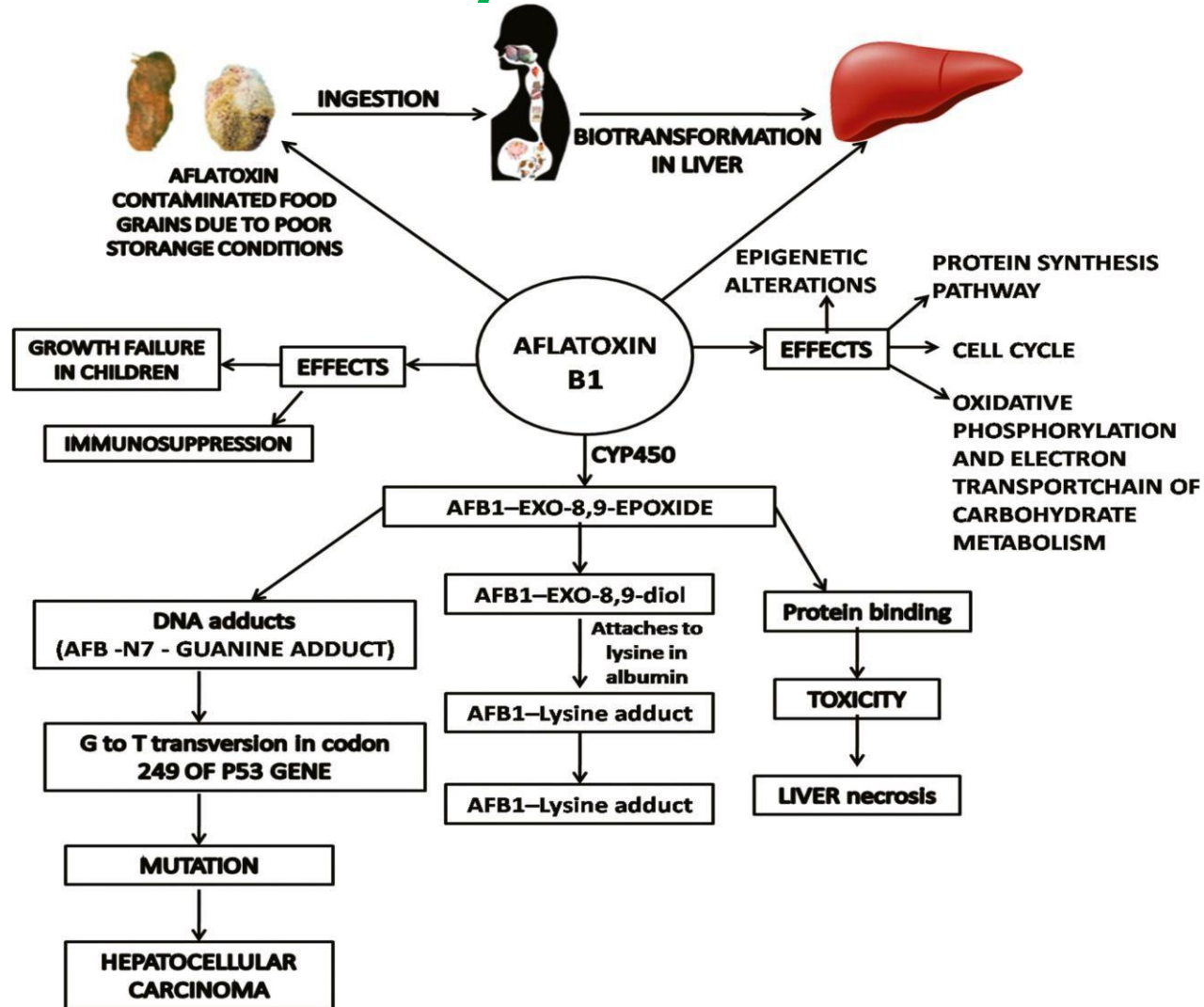


Fig. Aspergillus spore

Fungal carcinogens

Aflatoxin and Hepatocellular Carcinoma



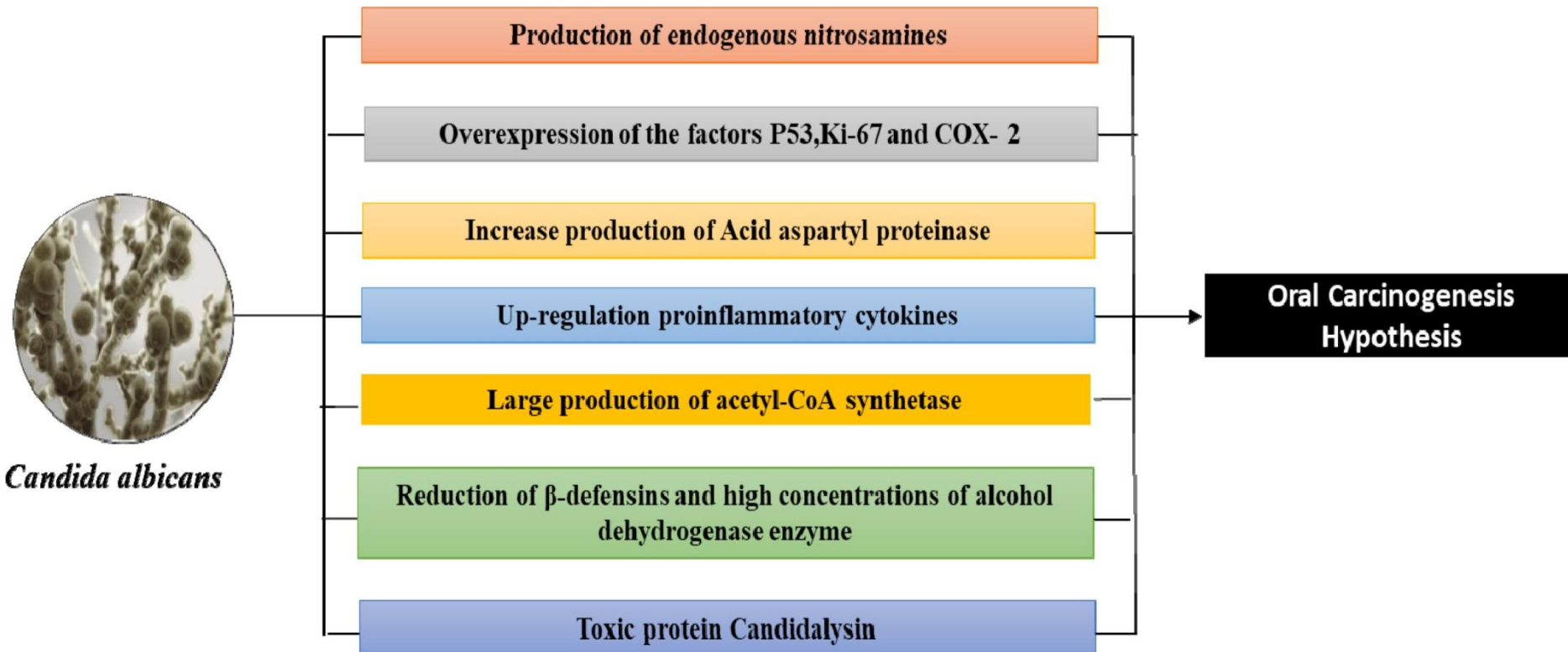
Fungal carcinogens

Candida spp.

- *Candida albicans* is a saprophytic fungus belonging to the Saccharomycetaceae family.
- It is found in the human microbiota of the gastrointestinal tract, oral cavity, and vagina.
- Candida genus can cause a wide range of pathologies.
- Oral candidiasis (with an incidence from 7% to 52%) has been observed in patients with neoplastic pathology (such as head and neck malignancies, hematopoietic neoplasms).

Fungal carcinogens

Candida spp.



Parasitic carcinogens

Parasitic pathogens	Endemic areas	Associated cancer	Proposed mechanism of carcinogenesis
Blood flukes			
Schistosoma haematobium	sSA	Urinary bladder cancer, adenocarcinoma, squamous cell carcinoma	Inflammation, oxidative stress caused by parasite-derived molecules
Schistosoma japonicum	sSA	Colorectal cancer, rectal cancer, squamous cell carcinoma, membranous nephropathy, metastatic lung cancer	Inflammation, oxidative stress caused by parasite-derived molecules
Schistosoma mansoni	sSA	Adenocarcinoma, colorectal cancer, hepatocellular carcinoma	Inflammation, oxidative stress caused by parasite-derived molecules
Liver flukes			
Opisthorchis viverrini	Southeast Asia	Cholangiocarcinoma	Inflammation, oxidative stress caused by parasite-derived molecules, cell proliferation, H. pylori mediated induction
Clonorchis sinensis	China, Korea, northern Vietnam	Cholangiocarcinoma	Inflammation, oxidative stress caused by parasite-derived molecules, cell proliferation

Parasitic carcinogens

Schistosoma haematobium

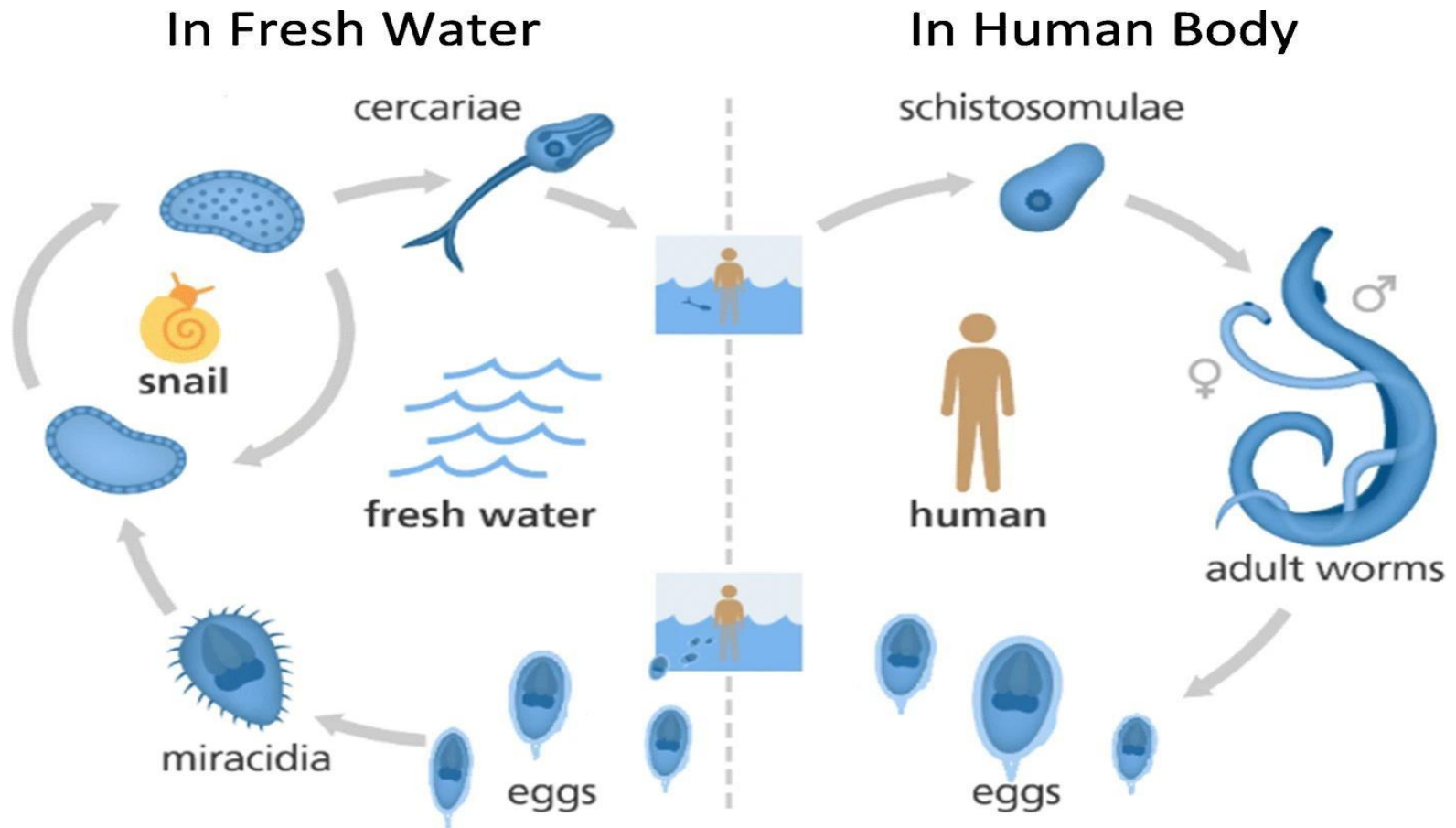


Fig. Life cycle of *Schistosoma haematobium* in freshwater (asexual reproduction in the snail) and in the human body (sexual reproduction in genitourinary venules) 12

Parasitic carcinogens

Schistosoma haematobium

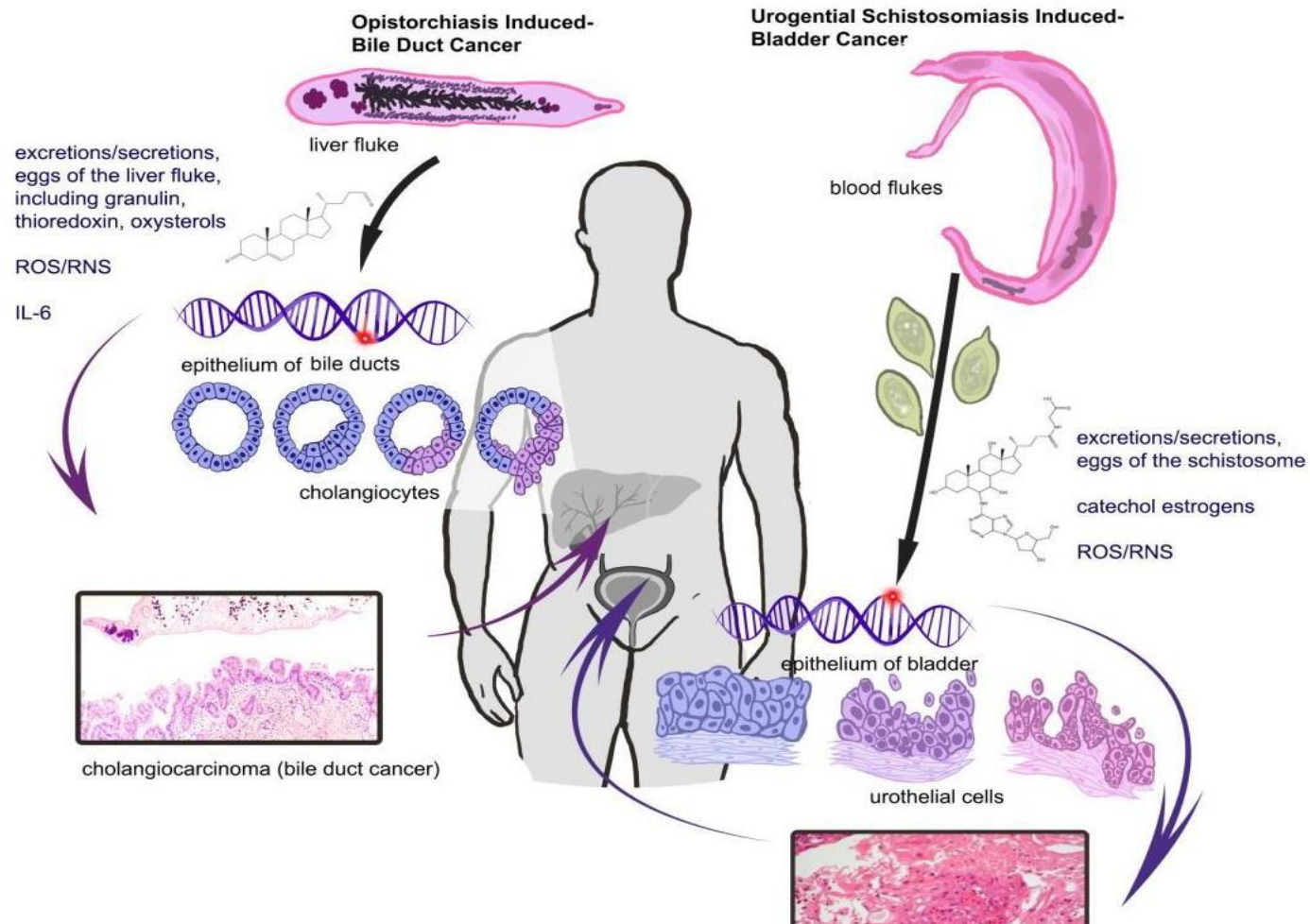
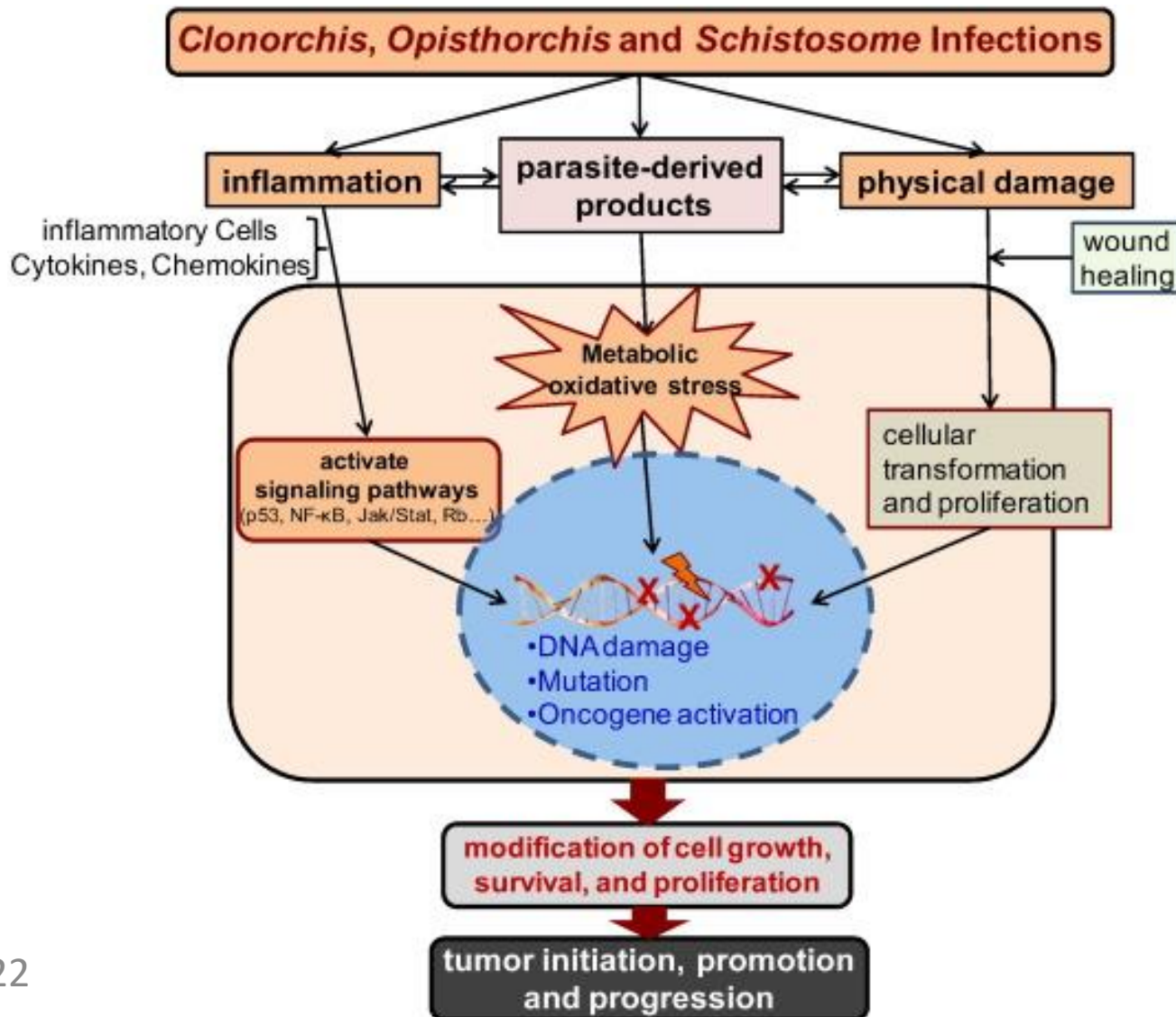
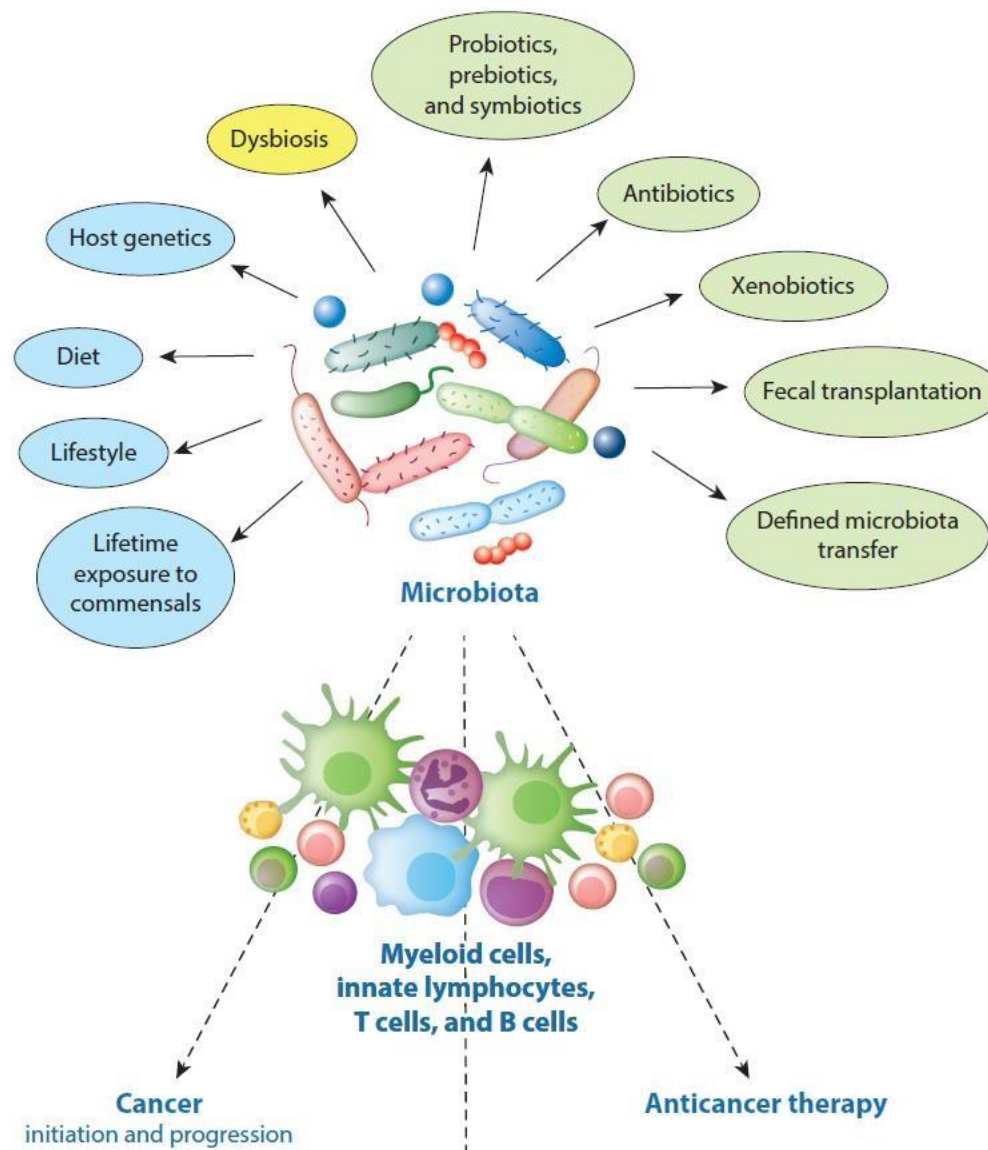


Fig. Schematic representation of cancers of the biliary and bladder/urogenital tract resulting from infection with *Opisthorchis viverrini* and *Schistosoma haematobium*

Mechanisms of carcinogenicity induced by infection with the liver and blood flukes



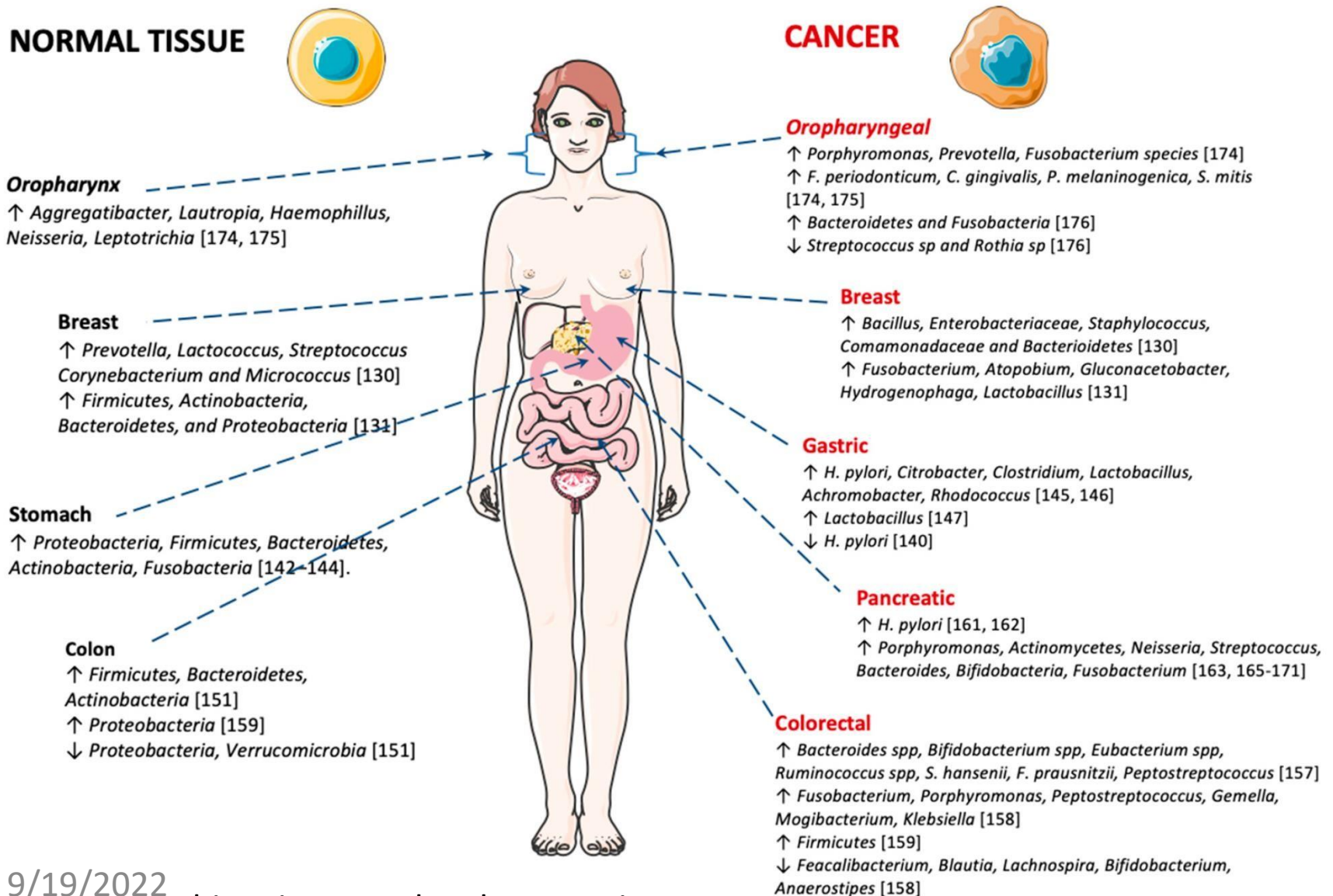
The Human Microbiome and Cancer



9/19/2022

Fig. The Human Microbiome and its Interaction with Tumor

The Human Microbiome and Cancer



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Fig. Microbiota in normal and cancer tissues

The End